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(Re)framing timber values: Cascade use metrics as a dynamic policy tool

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Abstract: Salvaged timber embodies a paradox of contemporary circularity: despite its capacity to store carbon and its long history of reuse, it remains marginal within modern construction. Its reuse is constrained by regulations that both shape and reflect the material's value. This paper reframes this paradox through the lens of value fluctuation–how financial, carbon, and social values oscillate and misalign across timber's successive lives, determining whether it circulates as resource, product, or waste. These sequential uses, known as cascade use, describe an approach to resource management that prioritizes biomass retention. Building on this framework, the paper proposes a value-centered cascade-use metric for salvaged timber as a speculative policy tool to visualize, modulate, and stabilize value, enabling longer and more successive cascades. Policy, thus understood as a design medium, can render the dynamic spatial and social consequences of material economies visible, diagnosing systemic frictions and prototyping regenerative, less harmful material futures.

Keywords: Reuse; Salvaged Timber; Policy; Cascade Use Metrics

1. Introduction

Salvaged timber embodies a paradox of contemporary circularity: despite its capacity to store carbon and its long cultural history of reuse, it remains marginal within modern construction (Cramer & Ridley-Ellis, 2020; Addis, 2006; Jarre et al., 2020). Recycled content in composite materials such as fiberboards or OSB boards are fairly common, followed by downcycled or reused finishes and furnishings, but directly reused timber components are limited mainly to historic structures where such reuse was once commonplace.

This paucity of reuse exists within the context of the global push towards decarbonization, where timber plays a significant role in policy and regulation. National and regional net-zero frameworks typically present increased timber use as a seemingly straightforward pathway to carbon mitigation, emphasizing wood's ability to sequester carbon during its lifespan (EESC, 2025; UK GOV, 2025). These policies valorize timber's capacity to store carbon, promoting greater virgin timber use and thus increased carbon storage. This logic assumes

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timber's value is exhausted once its initial carbon storage function has been fulfilled, privileging a single-use lifecycle. Short-term sequestration metrics prevail at the expense of longer-term reuse potential. This, and similar national and regional policies, determine the reuse outcomes of salvaged timber, both shaping and reflecting the material's value.

The notion of "value" in salvaged timber is highly complex and transcends simple economic value, consisting of many different forms, temporalities, and contingencies that often oscillate and misalign. This paper reframes salvaged timber through the lens of value fluctuation – how financial, carbon, social, and other values of salvaged timber rise, fall, and fail to cohere across timber's successive lives, determining whether it circulates as a resource, product, or waste. The sequential and multiple uses of timber, and the different forms it assumes in each use case, are described as cascade use: an approach to resource management that prioritizes extending the availability of biomass. The approach builds on insights from valuation studies, which emphasize that value is not intrinsic to objects but produced through socio-technical arrangements (Helgesson & Muniesa, 2013).

Cascade-use metrics are widely researched in the fields of circular economy, material resource management and regulation. Mantau's work on wood flow analysis (2015) utilizes quantitative metrics, including a "wood resource balance" and a "cascade factor" that calculates the "single or multiple use of a resource in a production system". Research by Jarre et al. (2020) highlights that implementation pathways to wood cascading are often vague and ambiguous, and crucial influencing factors such as policy limitations are frequently highlighted but remain barely investigated due to "substantial knowledge gaps" on political and socio-economic factors. Navare et al. (2022) analyze cascading wood use through the lens of GWP (global warming potential), a metric often overlooked in wood LCA. Their findings include that GWP decreases with increasing cascade steps and current carbon accounting may underestimate the climate benefit of cascading, thereby underestimating the secondary use of waste wood. Wang & Haller (2024) map Germany's circular wood flows with Sankey diagrams, illustrating how harvested wood moves through sawmills, construction, paper, and energy sectors. By visualizing material stocks and flows, such analysis pinpoints critical moments of intervention in the cascade trajectory. However, this analysis does not tie the material flow to value fluctuations that drive decision-making. Building on this existing work, the paper proposes a new, value-centered cascade-use metric for salvaged timber as a speculative policy tool capable of visualizing and modulating value fluctuations. Such a tool aims to bridge cascade-use metrics with value-based drivers that can guide policy and decision-making, providing a dynamic and adaptable base to understand possible interventions and outcomes.

This work positions policy itself as a design medium: a regulatory constraint but also an active site where representational design tools and methods can operate to shape complex and dynamic processes, as well as make visible and comprehensible the spatial and social consequences of material economies, facilitating negotiation, coordination, and recalibration. Seeking to go beyond traditional approaches that see policy as a rigid infrastructure, the proposed approach envisions policy-making as a malleable terrain for representational tools —diagrams, metrics— to render complex processes legible and reconfigurable. Engaging policy as a design medium is not only key to understanding the mechanisms that generate the built environment but also act upon it. Developing design methods and tools—like the cascade-use metric proposed here—that operate within policy's

own logics shall simultaneously reveal and reshape those logics' spatial and social consequences.

This approach is not unprecedented and there is a genealogy of interventions where representational tools and metrics have actively shaped regulatory frameworks governing material flows to lean upon. A particularly salient and useful example is the European Union's waste hierarchy, formalized in 1975 and updated in 2008 (EU 2008/98/EC). An inverted pyramid showing waste management options that also describe a value system, the hierarchy's visual representation has become a design tool deployed across scales: informing product design decisions, structuring municipal waste systems, shaping Extended Producer Responsibility (EPR) schemes. Designers and policymakers have iteratively refined the hierarchy's categories (adding "prepare for reuse" in 2008, debating the position of energy recovery), recognizing that each taxonomic choice redistributes material flows and economic incentives. The hierarchy functions as what we might call a "policy diagram"—a representational device that simultaneously describes and prescribes material futures. More recently, material passports and digital building logbooks piloted across EU contexts demonstrate architects and material scientists actively designing database architectures—determining what information fields to include, how to visualize material flows, who can access data—recognizing that database design is policy design with decades-long consequences for deconstruction practices and reuse markets.

These precedents reveal that design and representational methods have repeatedly operated through policy to render material realities visible, measurable, and therefore governable, establishing a trajectory the proposed cascade-use metric extends into salvaged timber's regulatory terrain.

By mapping the trajectories of multiple forms of value across multiple lifecycles, the cascade-use metric tool introduced in this paper can make visible where mismatched or unrecognized values might lead to material being incinerated rather than reused. The tool suggests that addressing these mismatches through targeted policy interventions could “flatten” or stabilize the destructive volatility of value, allowing reclaimed timber to be reused for more successive cascades and over more extended time periods. In developing the proposed cascade-use metric tool, illustrated through “zoom-ins” of specific timber-reuse scenarios and their consequences, the study aims to demonstrate how design informed by speculative policy tooling can both diagnose systemic frictions and prototype actionable, regenerative, and less harmful material futures.

2. Value fluctuations in salvaged timber

Within current European (including Swiss and Scandinavian) Circular Economy frameworks, the value of salvaged timber is defined through metrics that can be readily measured, monetized, or reported, primarily focusing on carbon storage and economic value. Policies such as the 2020 EU Circular Economy Action Plan emphasize the use of wood in construction for carbon storage, without distinguishing between new and reused wood (EU COM/2020/98). Other policies such as the forthcoming EU Construction Products Regulation (EU 2024/3110) leverage LCA (Life Cycle Assessment) and embodied carbon accounting, which similarly emphasize the immediate carbon footprint over carbon sequestration over

an extended period of time. (Pomponi & Moncaster, 2017) call for a shift in Circular Economy policy in the built environment from short-term manufactured products to buildings viewed as complex systems that shift over time and exist outside of the logic of manufacturing. Recent work has similarly argued that life-cycle assessment frameworks often fail to account for the multiple temporalities and relational consequences of reuse, decay, and other material afterlives (Deng et al., 2025). Under current LCA metrics, which do not account for extended carbon sequestration beyond the assessment period and do not distinguish reclaimed wood from new wood, wood is still viewed through the lens of short-term, single-use manufacturing.

Economic valuation, in the form of market protection regulations, taxonomy alignment and profitability, as seen in the EU Taxonomy Delegated Act (EC 2021/2178), privileges the efficiency, reliability, and predictability of material streams. EU waste recovery policies display a strong bias towards market-friendly metrics, favoring higher purity and consistency in recyclable materials streams. Such policy encourages circular economies to behave more akin to linear ones (Gregson et al., 2015). Economic and carbon value, both calculable and definable within market logics, determine the role of timber as a commodity within economies seeking to become circular, while marginalizing other, less quantifiable forms of value that perhaps belie this process.

In contrast to economic value and carbon-storage value, social, cultural, or labor values often remain excluded from official metrics, even though they shape material outcomes decisively. These diffuse forms of value are enacted through factors that are inherently more difficult to capture, such as embodied local knowledge and knowledge transfer, craftsmanship, local cultural identity, collective behavior, awareness, and participation in reuse infrastructures and economies. They are also shaped by questions of provenance, evidentiary detail, and how materials' political economies are rendered legible or ignored within architectural and regulatory systems (Malterre-Barthes, 2025). Labor invested in deconstruction, selective dismantling, or the processing of salvaged, non-standard timber often exceeds the explicit calculations of "economic value" pertaining to the material, yet has complex consequences both in terms of the cost of labor, time spent, and the added worth of local economic compensation and upskilling of the workforce. Cultural value, as heritage and vernacular value, has behavioral consequences that can generate and sustain reuse behavior far beyond what market movements and logics could allow.

Recent work exemplifies efforts towards expanding what counts as value in broader sustainability policy, such as the Nature Futures Framework (NFF) addressing land use and biodiversity (Dou et al, 2023). Developed under IPBES (Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services,) the NFF distinguishes between "Nature for Nature," "Nature for Society," and "Nature as Culture," reflecting an intrinsic and relational range of values that extends beyond market value.

The current value system of salvaged timber embodies an implicit hierarchy where quantifiable economic and carbon metrics dominate, while social, cultural, temporal, and labor values remain invisible and limited in their capacity to influence material outcomes. The recognition and integration of these plural and enmeshed values are critical to understanding, and thus acting upon, the value-driven relationships and factors that determine whether salvaged timber remains in circulation or becomes waste.

Across Europe, timber reuse is governed by an assemblage of codes, directives, standards, and voluntary frameworks. While diverse in function, these instruments consistently privilege calculable forms of value, such as carbon intensity, purity of material streams, conformity with standardized performance metrics, over the social, cultural, and labor inputs that often determine where reuse is practically achievable. The EU Construction Products Regulation (EU 2024/3110), remains critically influential, establishing certification and performance declarations that seek to create benchmarks for reliable and conformant material expectations, yet simultaneously constrain other forms of certification as well as uncertified or difficult-to-certify salvaged timber from circulation.

The EU Waste Framework Directive (EU 2008/98/EC) defines when materials are classified as waste or by-products: this is a critical juncture at which salvaged timber is either circulated or destined for incineration. The EU Taxonomy for Sustainable Activities (EU 2020/852) links reuse and carbon retention to financial valuation mechanisms through the regulation of access to "green" finance and investment based on quantifiable performance metrics emphasizing longevity and carbon efficiency.

Voluntary frameworks, such as the EU Level(s) and Green Public Procurement (GPP) operationalize value as a notion through reportable indicators, such as carbon intensity, recyclability, life-cycle cost, while leaving social value and labor input as qualitatively framed or absent. At the national scale, the Swiss waste law (VVEA 2015) and MuKEN cantonal harmonized energy regulations govern the hierarchies of construction and demolition waste, where waste prevention and energy recovery are primary values. Minergie, a voluntary energy standard, adds comfort and health metrics as values. The Swiss Wood Resource Policy (FOEN, n.d.) explicitly cites "keeping wood in circulation" as a goal, while the Swiss Parliamentary Initiative 20.433 "Strengthening the Swiss Circular Economy" positions reuse as a critical component of reaching decarbonization. Nevertheless, there is no harmonized policy or criteria outlining the reuse of wood as a resource. Technical certification regimes and grading standards govern whether reclaimed timber can re-enter structural markets, often creating barriers despite material viability (Falk, Cramer & Evans, 2012; Bergsagel & Heisel, 2023). The new Norwegian grading standard for reused timber, PR-NS 3691, addresses this barrier, outlining a path to grading through the visual appearance of timber (impurities such as impairment by moisture, decomposition, physical impact, surface treatment, fasteners) (Janjua et al, 2025).

Together, these regulatory instruments produce and enforce a hierarchy of values, where carbon capture, product, and market efficiency are explicit, calculable and actionable, while social, cultural, and labor contribution values remain implicit, incalculable, and excluded from metrics meant to determine cascade outcomes. The result is that the fluctuations of the more explicit forms of value (carbon, economic) are disconnected from, and cause volatility in, the more implicit forms of value (cultural, use, labor) that continue to fluctuate independently of the parameters that governance defines (Figure 1). Recognizing how the current landscape of governance both encodes and excludes certain forms of value in salvaged timber thus provides the basis for the development of visualization tools and methods that can display this relationship, pointing the way towards its recalibration across successive material cascades.

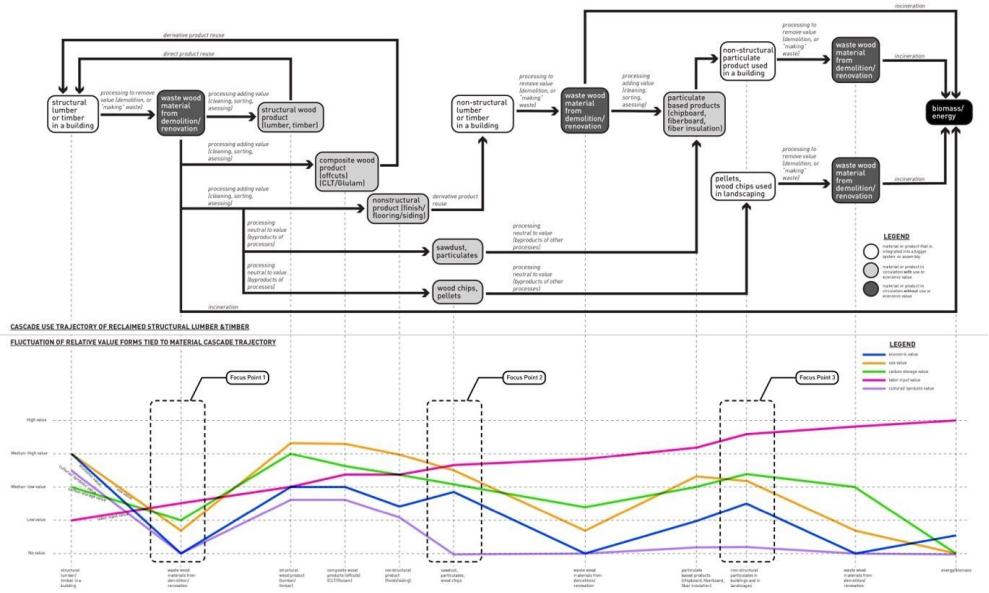


Figure 1 Cascade Use Trajectory and Value Fluctuations of Salvaged Timber (see Appendix for larger version)

3. The cascade-use and value visualization tool

3.1 Design of the tool

Designing a visualization tool to represent the relationship between cascade-uses and values requires the reconciliation of two parallel realities: first, that timber moves through multiple possible lifecycles, each shaped by technical, social, and regulatory constraints, and second, that existing policy tools overwhelmingly treat these trajectories as single-use, linear sequences. Modeling cascades, and building them into a policy-oriented visualization tool, is possible because timber's successive uses are already unfolding within overlapping allocation systems, such as carbon accounting frameworks, waste hierarchies, material certification regimes, and labor markets, which determine where value appears and where it disappears. To design a tool to visualize value fluctuations across cascade-uses is thus an intervention in these allocation systems, demonstrating how alternative forms of valuation could extend material lifespans, stabilize volatility, and enable continued reuse.

Current policy frameworks treat carbon storage as a one-time event rather than a long-term, dynamic process. Carbon-accounting conventions (as embedded in LCA, CPR declarations, and EU Taxonomy screening criteria) typically assess timber's value at the point of manufacture or first use, after which the material enters a zone of diminishing visibility. Subsequent cascades, across deconstruction or demolition, processing, remanufacturing, and reuse, fall outside of the scope of many regulatory metrics, which results in the paradox that salvaged timber, despite its reuse viability and carbon potential, is often treated as a waste or biomass product. Similar limitations characterize volume-based waste management systems, which evaluate timber through tonnages and cubic volumes. These frameworks render invisible the critical social, cultural, and labor-intensive forms of value that cannot be expressed within existing reporting structures.

In response, the cascade-use metric and value visualization tool introduced here as an early prototype proposes an alternative, dynamic, and context-adaptive framework that models timber's multiple successive lives and the fluctuating values that shape them. Rather than assessing timber at a single point in time, the tool traces how diverse forms of value rise, decline, or converge across cascade uses. These trajectories are neither static nor linear. They are influenced by external drivers such as regulatory changes, market conditions, social practices, and technological developments. By representing value as the result of dynamic interactions rather than fixed attributes, the tool reframes cascade-use modeling as an inquiry into how different forms of value co-produce material outcomes.

3.2 Derivation of value curves

The value curves presented in this paper are conceptual and heuristic representations, rather than empirically calibrated quantitative models. Their shape and relative positioning are derived from two sources: first, from preliminary qualitative insights gathered through interviews and discussions with actors involved in timber reuse, including architects, engineers, researchers, deconstruction and reuse practitioners, salvaged timber specialists, and policymakers; and second, from literature on wood cascading, circular construction, grading, deconstruction, and reuse governance, which documents recurring tensions between economic, environmental, technical, labor, and cultural value regimes.

The curves should therefore be read as ordinal and relational visualizations: they indicate the relative rise, decline, and misalignment of value forms across successive lifecycles rather than absolute or commensurable measurements. They synthesize how different stakeholder groups tend to prioritize and recognize value at different moments in the cascade. Future iterations of the tool would refine these trajectories through broader stakeholder participation, comparative case analysis, and participatory calibration. The current paper models a prototype single composite visualization, but future work may generate stakeholder-specific curve sets for different actors in the value chain.

3.3 Visualization of value fluctuations

Central to this approach is the visualization of value fluctuations. Each value curve– economic, carbon storage, labor input, and cultural value– operates as a time-based trajectory that decreases or increases depending on the material's condition, use case, regulatory environment, and socio-economic context. The value curves, when mapped together, reveal moments of volatility where one form of value collapses relative to others, placing the material at risk of exiting the cascade. The curves also reveal intervals of alignment where multiple values mutually reinforce one another, enabling the material to circulate more easily through successive uses. Stakeholders across the wood reuse chain evaluate timber through different criteria shaped by logistics, market incentives, and regulatory environments; the trajectories of value are therefore also stakeholder-dependent (Cramer & Ridley-Ellis 2020; Garcia & Hora 2017). Policy levers, addressed through this tool, can thus be reimagined not as static thresholds but rather as tools to modulate or "flatten" value volatility; for instance, raising certain value curves at key moments to prevent premature incineration.

As a visualization method, the cascade-use and value relationship represents how value fluctuations dictate possible cascade outcomes. Visualized iteratively, it has the potential to

simulate for stakeholders, including policymakers, designers, and researchers, how different interventions on materials can alter the shape of value trajectories, such as subsidies, certification frameworks, labor incentives, or design initiatives.

Future versions of the tool could incorporate computational modeling to run these scenarios dynamically, building on existing virgin timber frameworks allocating the resource to different products and parts of the economic cascade currently operating in the industry (Svilans and Tomsen, 2024; Gaudillière, et al., 2024) and predicting with the help of dedicated factors how specific policy interventions might stabilize or destabilize the curve across time.

A closer analysis of the movements of the value curve can reveal insights into voluntary and suffered fluctuations. Voluntary fluctuations refer to changes in value that can be intentionally influenced through policy, design, financial incentives, public programs or similar organizational decisions. Suffered fluctuations, by contrast, describe involuntary movements, such as market crashes, material degradation, contamination, or loss of cultural significance, that occur irrespective of intervention. The distinction, and the awareness of this distinction, matters since voluntary fluctuations signal opportunities for targeted action, whereas suffered fluctuations reveal limitations in meaningful action.

Three focus points, elucidated below, demonstrate possible scenarios where the visualization tool could positively impact cascade-use outcomes.

3.4 Focus point 1

Focus point 1 (Figure 2) visualizes a moment where the economic value curve falls below the carbon-storage curve.

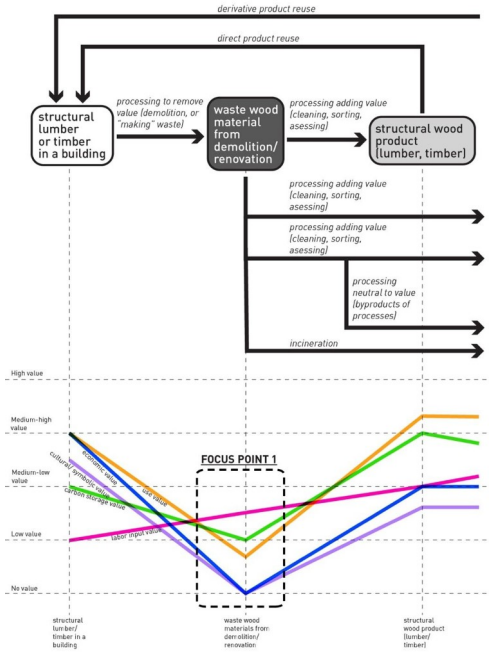


Figure 2 Focus Point 1: Critical intervention moment, risk of the material exiting the cascade.

At this juncture, the material is still highly viable for reuse from a carbon perspective. The stored carbon is still retained and could continue to be retained through subsequent cascades. If this juncture occurs within the first 50 years of the material's lifetime, this moment is extremely critical since incineration at this point would result in the timber being net carbon-positive (Hawkins, 2021). The economic value, in contrast, is low enough that demolition contractors, recyclers, or municipalities might choose incineration, resulting in an immediate release of carbon that could have been stored for decades longer. The risk emerges not from the materials' potential but from a misalignment of value curves. Here, a policy lever oriented towards carbon–such as a carbon-incentive payment for demonstrated carbon retention or a credit for avoided emissions–could raise the economic curve, stabilizing the fluctuation and delaying exit from the cascade.

3.5 Focus point 2

Focus point 2 (Figure 3) visualizes a moment where the economic value once again dips, relative to both the carbon value and the labor value.

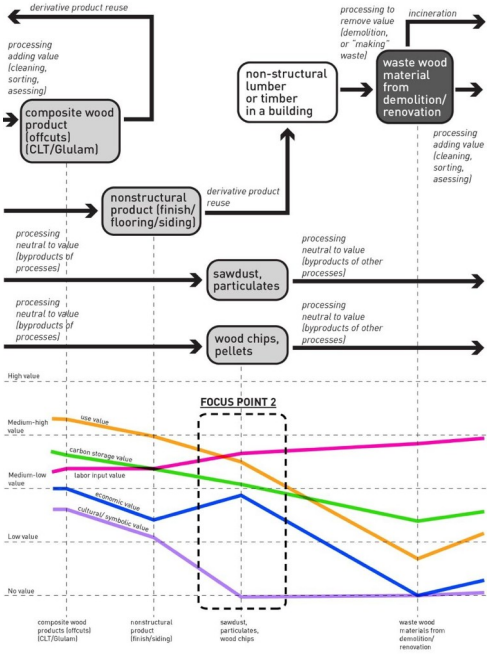


Figure 3 Focus Point 2: Critical intervention moment, risk of the material exiting the cascade.

The cumulative cost of labor-intensive processes such as selective deconstruction, manual sorting, or bespoke reprocessing can exceed the price that buyers are willing to pay for reclaimed or remanufactured timber products, resulting in downward pressure on the economic value (Diyamandoglu & Fortuna 2015). This is a moment for a critical intervention: targeted labor incentives, such as subsidies for deconstruction training, wage support for reuse-oriented labor, or fast-tracked certification programs that reduce administrative burdens, could raise the economic curve sufficiently to encourage the material to remain in circulation. Such interventions can act in tandem with carbon incentives, or provide a means to promote continued use where carbon incentives are no longer sufficient to ameliorate the economic value curve.

3.6 Focus point 3

Focus point 3 (Figure 4) visualizes a moment beyond which policy has limited ability to raise the declining value curves, due to an accumulation of suffered fluctuations.

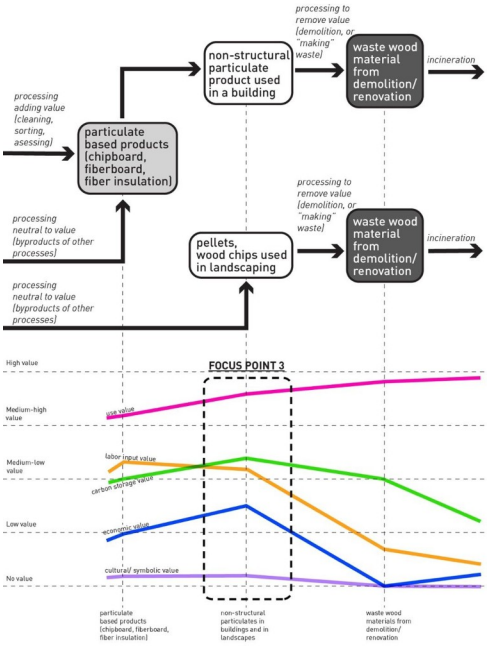


Figure 4 Focus Point 3: Where policy has limited application.

At this stage, the labor curve remains higher than others because increasing and cumulative amounts of labor are required to both dismantle and reconstruct the material, even as all other forms of value diminish. Beyond this point, policy is unlikely to meaningfully elevate the value curves, and the material is approaching the end of its viable cascades. Yet this does not imply that this threshold is entirely fixed. Upstream interventions, as will be discussed in the following section, have the potential to shift this trajectory upward, delaying or altering the timing of this limit.

3.7 Further development of the tool

Across these focus moments, the cascade-use metric and value visualization tool reveals where and how different interventions might extend material use and prevent premature incineration. A further developed, dynamic simulation could additionally model the cumulative effects of multiple interventions, allowing policymakers to see how combinations of carbon incentives, labor programs or similar strategies can interact to reshape material trajectories. The tool thus has the potential to function not merely as a diagnostic measure, but as an active speculation framework for the reimagination of regenerative material pathways. As such, the cascade-use and value visualization tool seeks to open space for coordinated intervention across multiple disciplinary domains, bridging design with policy.

4. Discussion

The cascade-use metric introduced in this paper foregrounds value as a dynamic and relational condition, rather than a fixed attribute of salvaged timber. The tool articulates how diverse forms of value fluctuate across successive uses, demonstrating that material outcomes are mediated by multiple social and political infrastructures that selectively recognize certain forms of value while sidelining others. The visualizations presented in Part 3 illustrate how targeted interventions could stabilize, or add time to periods of alignment between value forms, enabling timber to circulate over additional cascades before reaching its end point in incineration.

While Focus point 3 is described as a turning point, where policy interventions cannot stabilize, or recover, the declining curves of economic, labor, carbon, and cultural value, it is possible for design interventions to still make a stabilizing impact. Surveys of reclaimed wood users highlight the importance of cultural perceptions in shaping demand for reused timber products (Craig, Montague & Shmulsky 2024). If, for example, the intentional incorporation of design decisions into particulate-based products made of salvaged wood components can significantly contribute to the cultural and symbolic value of the building element, the increase in this value curve could “boost” the other curves – most significantly, the economic value curve. In this case, the material would more likely remain in use for longer, possibly even being reused as a particulate-based product for one more cascade cycle. This example underlines that policy and design interventions can complement one another in influencing material outcomes.

A key implication of this approach is the repositioning of designers, architects, researchers and community practitioners as actors with agency within the policy domain. Treating policy as a design medium reframes design practice away from solely the scale of the building or the object, towards the shaping of infrastructures of material valuation. Designers, beyond form-making, can negotiate the conditions that allow materials to appear, endure, or disappear. The cascade-use metric is a tool for expanding disciplinary agency: it renders the underlying value dynamics legible enough to deliberate upon, contest, and reconfigure. In this sense, design decisions and policy decisions are mutually influential, in that assumptions in one domain can influence possibilities in the other.

Efforts to stabilize value volatility pose several risks: value forms that resist quantification, such as cultural value and labor input, may become instrumentalized; distinct yet similar value forms may be condensed into reductive metrics. Stakeholders may experience the effects of value rebalancing unevenly: laborers may face uneven demand and compensation. Regulators may encounter administrative and organizational complexity. These risks signal the need for longer-term calibration of the proposed metrics to local contexts and applications.

5. Conclusion

Salvaged timber carries urgent potential to interrupt some of construction's most wasteful practices. Reusable timber—already harvested, already processed, already storing carbon—is systematically incinerated or downcycled rather than recirculated (Figure 5 below, illustrating possible cascade uses within a single timber-frame building). This squandering

occurs not due to material limitations but through regulatory frameworks that fail to recognize or stabilize the multiple forms of value timber embodies across its successive lives. Addressing this requires more than technical innovation in joinery or grading standards; it demands interrogating and reshaping the policy infrastructures that determine whether materials circulate or become waste.

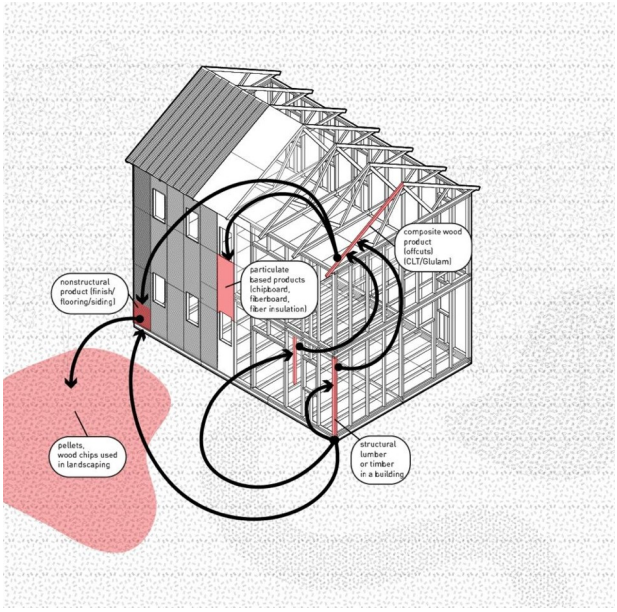


Figure 5 Cascade uses of wood products, illustrated cycles within a single building

Because of this, the present work seeks to position salvaged timber as both an urgent case study and methodological lens—a material whose successive transformations across regulatory, economic, and social systems reveal how value itself is co-produced, how it fluctuates, and how these fluctuations determine material futures. It is worth noting that the chosen value types and corresponding fluctuations rely on the authors' judgment for the scope of this paper – further iterations of the visualization tool could be expanded to include further forms of value, depending on the tool user's perspective.

The visualization framework demonstrated here is a conceptual prototype, a speculative design instrument aimed at structuring policy conversations around material reuse. Its primary audience includes policymakers, researchers, designers, and practitioners involved in circular construction. In this sense, the tool functions as a boundary object capable of supporting dialogue between actors who operate under different valuation frameworks. Future development of the tool would benefit from participatory workshops in which stakeholders collectively map and adjust value trajectories based on their own experiences and priorities. Such co-development processes could enable the visualization to evolve into a shared policy design device capable of integrating empirical data and practitioner knowledge.

The cascade-use metric tool proposed here demonstrates that policy can become an active site for design experimentation: a terrain where visualization, diagramming, and metric-making operate as methods to diagnose systemic frictions, make value dynamics legible, and prototype interventions that could stabilize destructive volatility. Additionally, what begins with salvaged timber extends outward. Every material transition confronts similar regulatory

inadequacies and faces policies designed for stable commodities applied to dynamic cascades, frameworks privileging economic metrics while excluding the labor, knowledge, and cultural practices that determine whether circularity succeeds or fails. The speculative policy approach modeled here—treating visualization as intervention, treating metrics as negotiable rather than neutral—becomes increasingly necessary as material economies are forced to shift.

Yet the method alone is insufficient without confronting who designs these policy tools and toward what ends. The cascade-use metric proposed here has emerged from academic research with access to regulatory knowledge, data visualization capacity, and institutional platforms for dissemination. Salvage workers, community repair networks, and informal reuse economies—those whose practices already enact circular logics that policy claims to promote—rarely have equivalent access to shape the metrics that govern their work. Speculative policy design can thus actively redistribute the capacity to design regulatory futures: opening policy-making processes to those with situated material knowledge, creating accessible tools for visualizing and contesting value hierarchies, legitimizing forms of expertise that regulatory systems currently exclude. This means that other tools will be needed: participatory platforms for mapping local material flows, adaptive frameworks that evolve with changing ecologies and economies, regulatory languages flexible enough to accommodate regional variation without abandoning coordination across scales. Designers, architects, and planners have roles to play not as authors of universal solutions but as facilitators of collective tool-making, translating between technical possibility and political negotiation.

Salvaged timber will continue circulating through construction sites, salvage yards, and incinerators. Whether it cascades through successive reuses or burns after single use depends less on inherent physical and material properties than on whether regulatory systems can recognize and stabilize the forms of value that make reuse viable. Policy, redesigned as dynamic infrastructure rather than static rule, could enable those longer cascades. At the same time, it is crucial to acknowledge that technical metrics traffic in politics, making claims about what is visible, valuable, and governable. Visualization may privilege certain knowledges while excluding others, and every diagram of material and value flows is a diagram of power. Recognizing this does not diminish the utility, design, and research value of the cascade-use visualization framework; rather, it underscores the stakes of designing such tools and methods with care, reflexivity, and cross-disciplinary collectivization.

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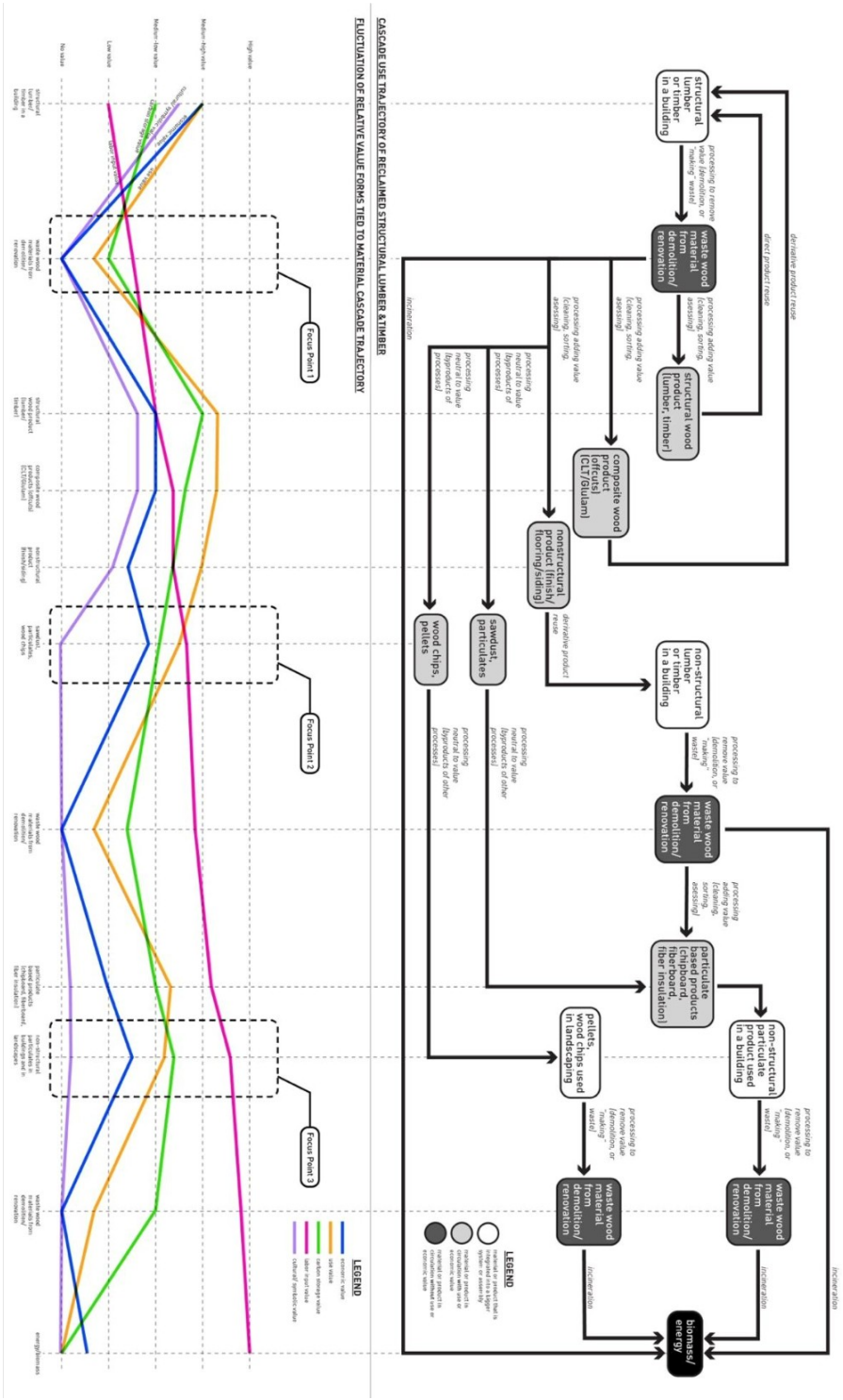
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7. Appendix


CASCADE USE TRAJECTORY OF RECLAIMED STRUCTURAL LUMBER &TIMBER
FLUCTUATION OF RELATIVE VALUE FORMS TIED TO MATERIAL CASCADE TRAJECTORY
structural lumber or timber in a building
waste wood material from demolition/ renovation
structural wood product (lumber, timber)
composite wood product (CLT/Glulam)
nonstructural product (finish/ flooring/siding)
sawdust, particulates
wood chips, pellets
non-structural lumber or timber in a building
particulate based products (chipboard, fiberboard, fiber insulation)
non-structural particulate product in a building
pellets, wood chips used in landscaping
biomass/ energy
direct product reuse
derivative product reuse
incineration
LEGEND
Focus Point 1
Focus Point 2
Focus Point 3
LEGEND